BTS-III-11.18-1718 Reg. No. | A

B.Tech. Degree III Semester Examination November 2018

ME 15-1303 MECHANICS OF SOLIDS
(2015 Scheme)

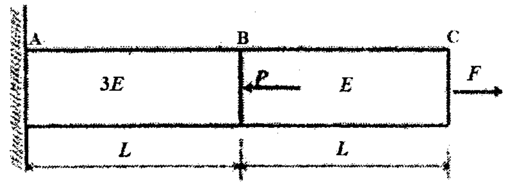
Time: 3 Hours Maximum Marks: 60

PART A
(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) Derive the relationship connecting the Young's modulus, bulk modulus and Poisson's ratio for a homogeneous isotropic material.

(b) A horizontal bar with a constant cross section is subjected to loading as shown in the figure. The Young's moduli for the sections AB and BC are 3E and E respectively. What is the ratio P/F for the deflection at C to be zero?

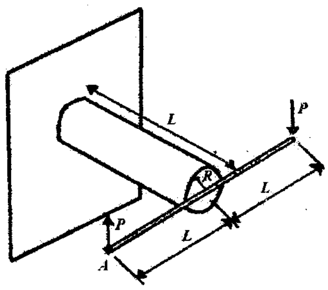


(c) A thin walled spherical shell is subjected to an internal pressure. If the radius of the shell is increased by 1% and the thickness is reduced by 1%, with the internal pressure remaining the same, what is the change in the circumferential (hoop) stress?

(d) For a loaded cantilever beam of uniform cross section the bending moment in N.mm along the length is given by $M(x) = 5x^2 + 10x$, where x is the distance in mm measured from the free end. What is the magnitude of the shear force in N at x = 10 mm?

(e) Derive the torsion equation $\frac{T}{J} = \frac{G\theta}{l} = \frac{\tau}{r}$ for a circular shaft subjected to a torque T.

(f) A rigid horizontal rod of length 2L is fixed to a cylinder of radius R as shown in the figure. Vertical forces of magnitude P are applied at the two ends as shown. The shear modulus for the cylinder is G. What is the vertical displacement of end A?

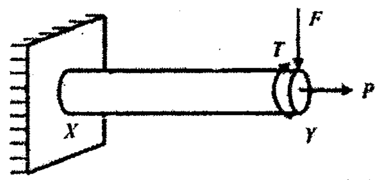


(g) A cantilever beam having square cross section of side a is subjected to an end load. If a is increased by 19%, what is the percentage change in the tip deflection of the beam?

(h) The cross sections of two solid bars made of the same material are square and circular. The square cross section has flexural rigidity K_1, while the circular cross section has flexural rigidity K_2. Both sections have same cross sectional area. What is the ratio K_1/K_2?

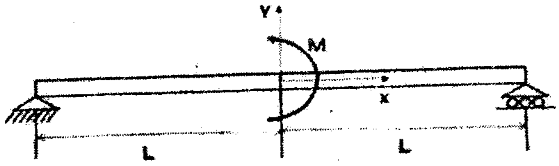
(i) A machine element XY fixed at end X is subjected to an axial load P, transverse load F, and twisting moment T at its free end. Which of the following is the most critical point from the strength point of view? Justify you answer.

(i) A point on the circumference at location Y.
(ii) A point at the center at location Y.
(iii) A point on the circumference at location X.
(iv) A point at the center at location X.



(P.T.O.)

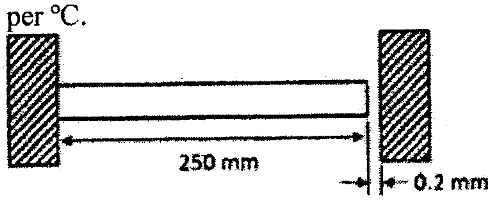
(j) A simply supported beam of length 2L is subjected to a moment M at the mid point $x = 0$ as shown in the figure. The deflection in the domain $0 \le x \le L$ is given by $y = \frac{-Mx}{12EIL}(L-x)(x+C)$ where E is the Young's modulus, I is the area moment of inertia of the cross section of the beam about the neutral axis, and C is a constant which is to be evaluated. What is the slope at the center $x = 0$?



PART B

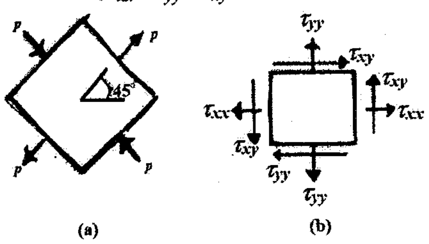
$(4 \times 10 = 40)$

II. A circular metallic rod of length 250 mm is placed between two rigid immovable walls as shown in the figure. The rod is in perfect contact with the wall on the left side and there is a gap of 0.2 mm between the rod and the wall on the right side. If the temperature of the rod is increased by 200°C, what is the axial stress developed in the rod? Take the Young's modulus of the material of the rod as 200 GPa and the coefficient of thermal expansion as 10^{-5} per °C.



OR

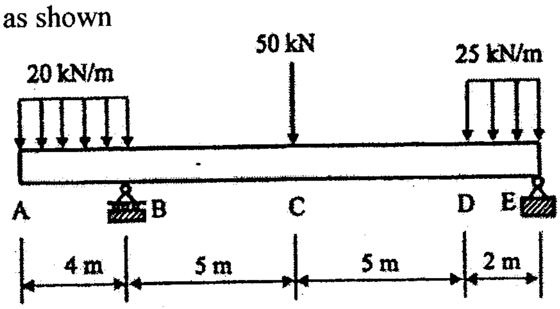
III. The state of stress at a point on an element is shown in figure (a). The same state of stress is shown in another co-ordinate system in figure (b). Determine the components $(\tau_{xx}, \tau_{yy}, \tau_{xy})$ if the value of $p = 10$ MPa



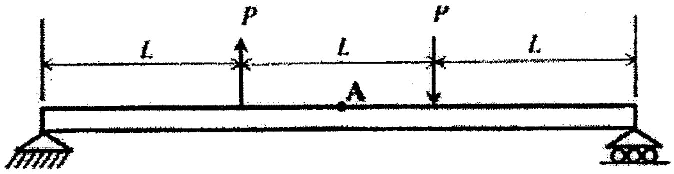
IV. A propeller shaft of a small ship is made of a solid steel bar 100 mm in diameter. The allowable stress in shear is 50 MPa and the allowable angle of twist is 1° in 1.5 m length. Assuming the shear modulus of the material of the shaft is 80 GPa, determine the maximum torque that can be applied to the shaft.

OR

V. Draw the shear force and bending moment diagrams for the beam loaded and supported as shown

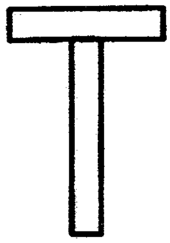


VI. A simply supported beam of length 3L is subjected to the loading as shown in the figure. It is given that P = 1 kN, L = 1 m, Young's modulus E = 200 GPa. The cross section is square with dimension 10 mm × 10 mm. Determine the bending stress at the point A located at the top surface of the beam at a distance 1.5 L from the left end.



OR

VII. Determine the shear stress distribution across the cross section of a beam having shape given in figure for a shear force V = 10 kN. The dimensions of the flange (horizontal) is 400 mm × 10 mm and that of the web (vertical) is 700 mm × 10 mm.



VIII. Determine the equation of the deflection curve for a cantilever beam of length L subjected to a uniformly distributed load of intensity q N/m. Also find the deflection and angle of rotation at the free end of the beam.

OR

IX. A pinned end column of aluminium with Young's modulus E = 73 GPa and length L = 2 m is constructed of circular tubing with outside diameter d = 50 mm. The column must resist an axial load P = 14 kN with a factor of safety n = 2 with respect to buckling. Determine the required thickness of the tube.

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